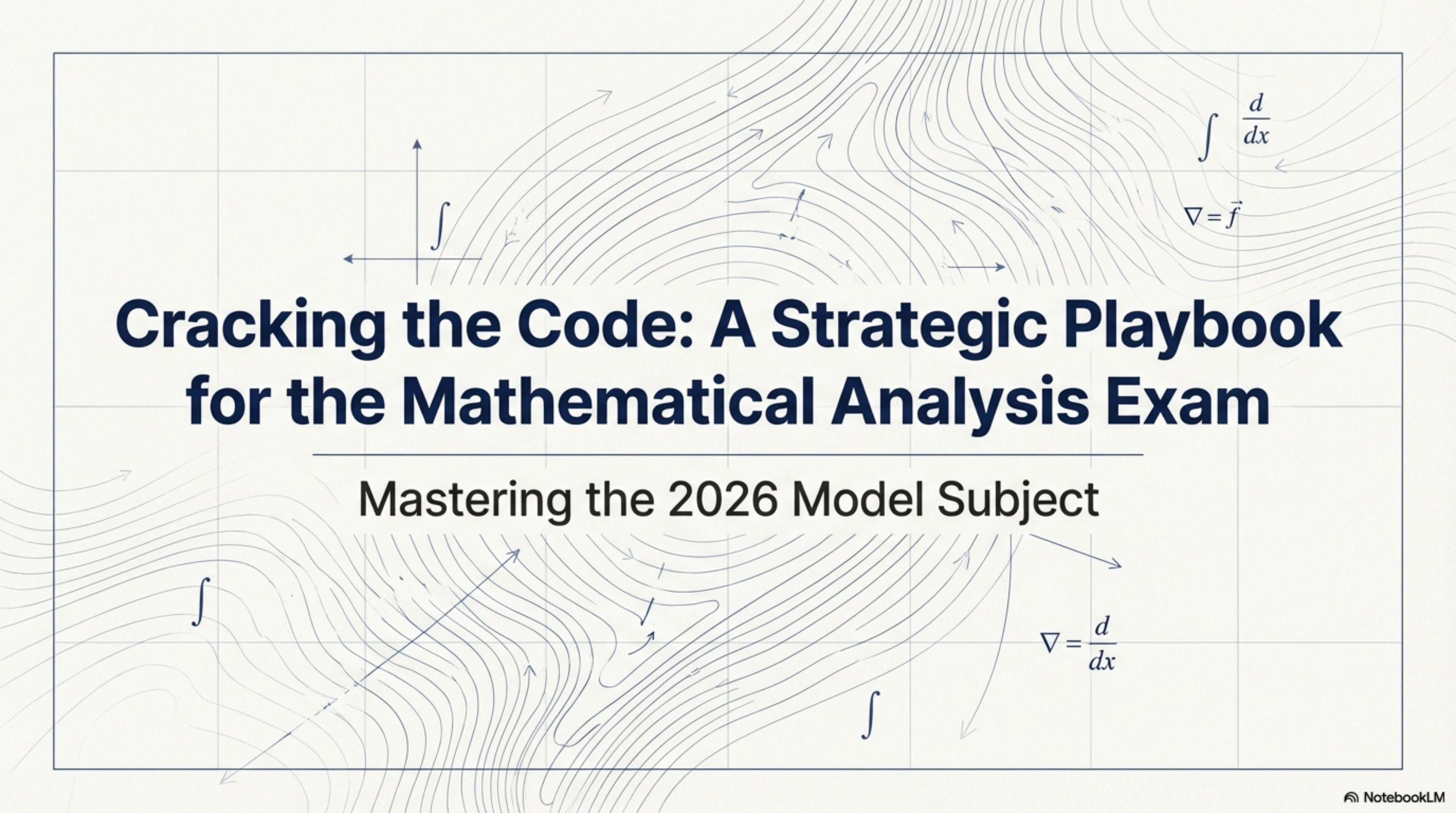


The background of the slide features a light gray grid overlaid with a complex vector field of thin, curved lines with arrows. Various mathematical symbols are scattered across the background: an integral symbol $\int$ is in the upper left; the expression $\int \frac{d}{dx}$ is in the upper right; the expression $\nabla = \vec{f}$ is in the middle right; another integral symbol $\int$ is in the lower left; and the expression $\nabla = \frac{d}{dx}$ is in the lower right. A horizontal line is positioned below the main title.

Cracking the Code: A Strategic Playbook for the Mathematical Analysis Exam

Mastering the 2026 Model Subject

The Exam Blueprint: Know the Terrain

Total Points: 10		
Exam Section (7 points)		Seminar Activity (2.0 points)
Problem 1 (4.0 points) The Foundations Gauntlet. <i>Strategic Note: High value, tests breadth of knowledge. Speed and accuracy are key.</i>		
Problem 2 (1.5 points) The Pattern-Recognition Challenge. <i>Strategic Note: Tests deeper problem-solving. Often involves a 'trick' or non-obvious identity.</i>		Granted ('Oficiu') (1.0 point)
Problem 3 (1.5 points) The Geometric Application. <i>Strategic Note: Tests the link between algebra and geometry. Visualizing the problem is the first step to solving it.</i>		

Our approach is not to just solve problems, but to develop a repeatable strategy for each *type* of problem you will face.

Problem 1 (4 Points): The Foundations Gauntlet

This section is a test of your command over four foundational pillars of analysis: limits by definition, geometric series, integral convergence, and multivariable gradients. Mastery here requires both precision and efficiency. Let's break down the playbook for each.

$$\text{a) } \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{n+1} = 0 \text{ (by definition)}$$

$$\text{b) } \sum_{n=1}^{\infty} \left(\frac{-1}{2} \right)^{3n+1}$$

$$\text{c) Convergence of } \int_0^1 (x \ln x)^2 dx$$

$$\text{d) } \nabla f \left(\frac{\pi}{2}, 2 \right) \text{ for } f(x, y) = 2 \ln \left(\tan \frac{x}{y} \right)$$

1a) The Blueprint: Mastering the ε - N Definition of a Limit

The Formal Definition

A sequence a_n converges to a limit L if, for every arbitrary small positive number ε , we can find a natural number N (which may depend on ε) such that for all $n > N$, the distance between a_n and L is less than ε .

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n > N, |a_n - L| < \varepsilon$$

- **Strategic Insight:** This isn't just a formula to memorize; it's a logical challenge. The goal is to algebraically manipulate the inequality $|a_n - L| < \varepsilon$ to isolate n and prove that such an N must exist.

1a) The Build: Applying the Definition

1. **Set up the inequality:** We need to show $\left| \frac{\sqrt{n}}{n+1} - 0 \right| < \varepsilon$, which simplifies to $\frac{\sqrt{n}}{n+1} < \varepsilon$.
2. **Find a simpler upper bound:** The key is to simplify the expression. We know $n+1 > n$. Therefore, $\frac{1}{n+1} < \frac{1}{n}$, which means $\frac{\sqrt{n}}{n+1} < \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}}$. The crucial inequality $\frac{\sqrt{n}}{n+1} < \frac{1}{\sqrt{n}}$.
3. **Solve the simpler inequality:** If we can make $\frac{1}{\sqrt{n}} < \varepsilon$, our original inequality will also hold. This gives $\sqrt{n} > \frac{1}{\varepsilon}$, so $n > \frac{1}{\varepsilon^2}$.
4. **Define N :** We have proven that for any $\varepsilon > 0$, if we choose N to be any integer greater than or equal to $\frac{1}{\varepsilon^2}$ (e.g., $N = \left\lceil \frac{1}{\varepsilon^2} \right\rceil$), then for all $n > N$, our condition holds. The limit is justified.

ε -N Strategy

1. Start with $|a_n - L| < \varepsilon$.
2. Find a simpler, larger expression to bound $|a_n - L|$. This is the most creative step.
3. Solve the simplified inequality for n in terms of ε .
4. Formally state your choice for N .

1b & 1c) The Blueprint: Pattern Recognition for Series and Integrals

Geometric Series (Problem 1b)

Concept: A geometric series has the form $\sum ar^n$.

Convergence Rule: It converges if and only if the common ratio $|r| < 1$.

Sum Formula: $S = \frac{a}{1-r}$ for an infinite series starting at $n=0$. (*Note: Be careful with the starting index n !*)

Improper Integral Convergence (Problem 1c)

Concept: An integral $\int_a^b f(x) dx$ is improper if $f(x)$ has an infinite discontinuity at a point in $[a, b]$.

Key Tool: The Comparison Test: If $0 \leq f(x) \leq g(x)$ and $\int g(x) dx$ converges, then $\int f(x) dx$ converges.

P-Integral Benchmark: $\int_0^1 \frac{1}{x^p} dx$ converges if $p < 1$.

1b & 1c) The Build: Rapid Execution

Problem 1b Solution

Rewrite the term: The series term can be rewritten as

$$\left(\left(-\frac{1}{2}\right)^{3n+1}\right) = \left(-\frac{1}{2}\right) \cdot \left(-\frac{1}{2}\right)^n = \left(-\frac{1}{2}\right) \cdot \left(-\frac{1}{8}\right)^n.$$

Identify parameters: This is a geometric series with a common ratio $r = -\frac{1}{8}$. Since the absolute value $|r| < 1$, the series converges.

Calculate sum: The first term of the series (at $n = 1$) is $a = \left(-\frac{1}{2}\right) \cdot \left(-\frac{1}{8}\right) = \frac{1}{16}$. The sum is given by the formula $\frac{a}{1-r}$. Substituting the values, we get

$$\frac{\frac{1}{16}}{1 - \left(-\frac{1}{8}\right)} = \frac{\frac{1}{16}}{\frac{9}{8}} = \frac{1}{18}.$$

Problem 1c Solution

Identify the issue: The integrand function $(x \ln x)^2$ is problematic as $x \rightarrow 0^+$, indicating a potential improper integral.

Analyze the limit: We know that $\lim_{x \rightarrow 0^+} x \ln x = 0$ (this can be shown using L'Hôpital's rule on $\frac{\ln(x)}{1/x}$). This result suggests the function is "well-behaved" and the integral likely converges.

Justify with Comparison: For $x \in (0, 1)$, we know that $|\ln x| < \frac{C}{x^\alpha}$ for any $\alpha > 0$. Let's choose $\alpha = \frac{1}{4}$. Then $(x \ln x)^2 < \left(x \cdot \frac{C}{x^{1/4}}\right)^2 = C^2(x^{3/4})^2 = C^2x^{3/2}$.

Conclude: Since the integral $\int_0^1 x^{3/2} dx$ converges (as it is a p-integral with $p = -\frac{3}{2}$, and $-\frac{3}{2} > -1$), our original integral converges by the Comparison Test.

The Playbook Summary:

For Series: Manipulate the term into a standard form (e.g., p-series, 'geometric').

For Integrals: Identify the point of discontinuity. Analyze the function's behavior near that point. Use limit comparison or direct comparison with a known p-integral.

1d) The Blueprint & Build: Calculating the Gradient

Concept Definition: The gradient of a scalar function $f(x, y)$, denoted ∇f , is a vector containing its partial derivatives: $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$. It points in the direction of the steepest ascent of the function.

Problem: $f(x, y) = 2 \ln\left(\tan\left(\frac{x}{y}\right)\right)$. Find $\nabla f(\pi/2, 2)$.

Calculation:

Step 1: Find $\frac{\partial f}{\partial x}$ (using the chain rule)

$$\frac{\partial f}{\partial x} = 2 \cdot \frac{1}{\tan(x/y)} \cdot \sec^2 \frac{x}{y} \cdot \frac{1}{y} = \frac{4}{y \cdot \sin(2x/y)}$$

$$\text{At } (\pi/2, 2): \frac{x}{y} = \frac{\pi}{4} \cdot \frac{\partial f}{\partial x} = \frac{4}{2 \sin(\pi/2)} = 2.$$

Step 2: Find $\frac{\partial f}{\partial y}$ (using the chain rule)

$$\frac{\partial f}{\partial y} = 2 \cdot \frac{1}{\tan(x/y)} \cdot \sec^2 \frac{x}{y} \cdot \left(\frac{-x}{y^2} \right) = \frac{-4x}{y^2 \cdot \sin(2x/y)}$$

$$\text{At } (\pi/2, 2): \frac{\partial f}{\partial y} = \frac{-4(\pi/2)}{4 \sin(\pi/2)} = \frac{-2\pi}{4} = -\frac{\pi}{2}.$$

Step 3: Assemble the Gradient Vector

$$\nabla f\left(\frac{\pi}{2}, 2\right) = \left(2, -\frac{\pi}{2}\right).$$

Playbook Note: For gradient problems, the process is mechanical: apply the chain rule carefully, substitute the point's coordinates, and assemble the final vector. The main risk is algebraic error.

Problem 2 (1.5 Points): The Telescoping Series Deep Dive

Problem Statement: Establish the nature of the series with positive terms and then calculate its sum:

$$\sum_{n=0}^{\infty} \operatorname{arctg} \left(\frac{2^n}{1 + 2^{2n+1}} \right)$$



Strategic Insight: When asked to find the sum of a complex-looking series, it is often not a geometric or p-series. Suspect a **telescoping series**. The goal is to rewrite the general term a_n as a difference, $a_n = b_{n+1} - b_n$.

Key Tool: The most relevant identity for arctan is:

$$\arctan(x) - \arctan(y) = \arctan\left(\frac{x-y}{1+xy}\right)$$

Unmasking the Telescoping Series: The Build

Step 1: Deconstruct the Term

Our goal is to match $\frac{2^n}{1+2^{2n+1}}$ with $\frac{x-y}{1+xy}$.

Let's rewrite the denominator: $1 + 2^{2n+1} = 1 + 2 \cdot 2^{2n} = 1 + 2 \cdot (2^n) \cdot (2^n)$.

Let's try $x = 2^{n+1}$ and $y = 2^n$. Then $xy = 2^{2n+1}$. And $x - y = 2^{n+1} - 2^n = 2^n(2 - 1) = 2^n$.

It's a perfect match! The general term a_n is $\arctan(2^{n+1}) - \arctan(2^n)$.

Step 2: Write Out the Partial Sum (S_N)

$$S_N = \sum_{n=0}^N [\arctan(2^{n+1}) - \arctan(2^n)]$$

$$S_N = (\arctan(2^1) - \arctan(2^0)) + (\arctan(2^2) - \arctan(2^1)) + \dots + (\arctan(2^{N+1}) - \arctan(2^N))$$

The intermediate terms cancel out, leaving: $S_N = \arctan(2^{N+1}) - \arctan(2^0) = \arctan(2^{N+1}) - \arctan(1)$.

Step 3: Calculate the Limit

The sum of the series is $S = \lim_{N \rightarrow \infty} S_N$. $S = \lim_{N \rightarrow \infty} [\arctan(2^{N+1}) - \frac{\pi}{4}] = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$.

Conclusion: The series converges, and its sum is $\frac{\pi}{4}$.

The Playbook: A Strategy for Hidden Telescoping Series

The Core Principle: If the problem asks for the exact sum of an unusual series, your primary suspect is a **telescoping series**.

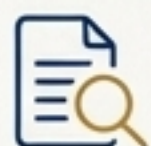
The Algorithm:



1. **Scan for Clues:** Look for functions where a difference identity exists (e.g., arctan, ln, partial fractions like $\frac{1}{n(n+1)}$).



2. **Match the Pattern:** Attempt to rewrite the general term a_n in the form $b_{(n+1)} - b_n$ or $b_n - b_{(n+1)}$ by applying the relevant identity. This is the main puzzle.



3. **Analyze the Partial Sum:** Write out the first few terms and the last term of the partial sum S_N to clearly see the cancellation pattern.



4. **Take the Limit:** Calculate $\lim_{N \rightarrow \infty} S_N$ to find the final sum. Watch for terms that go to zero, infinity, or a constant.

Common Identities to Memorize:

$$\arctan(x) - \arctan(y) = \arctan\left(\frac{x-y}{1+xy}\right) \quad \ln(a) - \ln(b) = \ln\left(\frac{a}{b}\right) \quad \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1} \text{ (Partial Fractions)}$$

Problem 3 (1.5 Points): Mastering Double Integrals

Problem Statement:

Calculate the double integral $\iint_A \frac{1}{(x+y)^2} dx dy$, where

$$A = \{ (x, y) \in \mathbb{R}^2 \mid x \geq 0, y \geq 0, 1 \leq x+y \leq 2 \}.$$

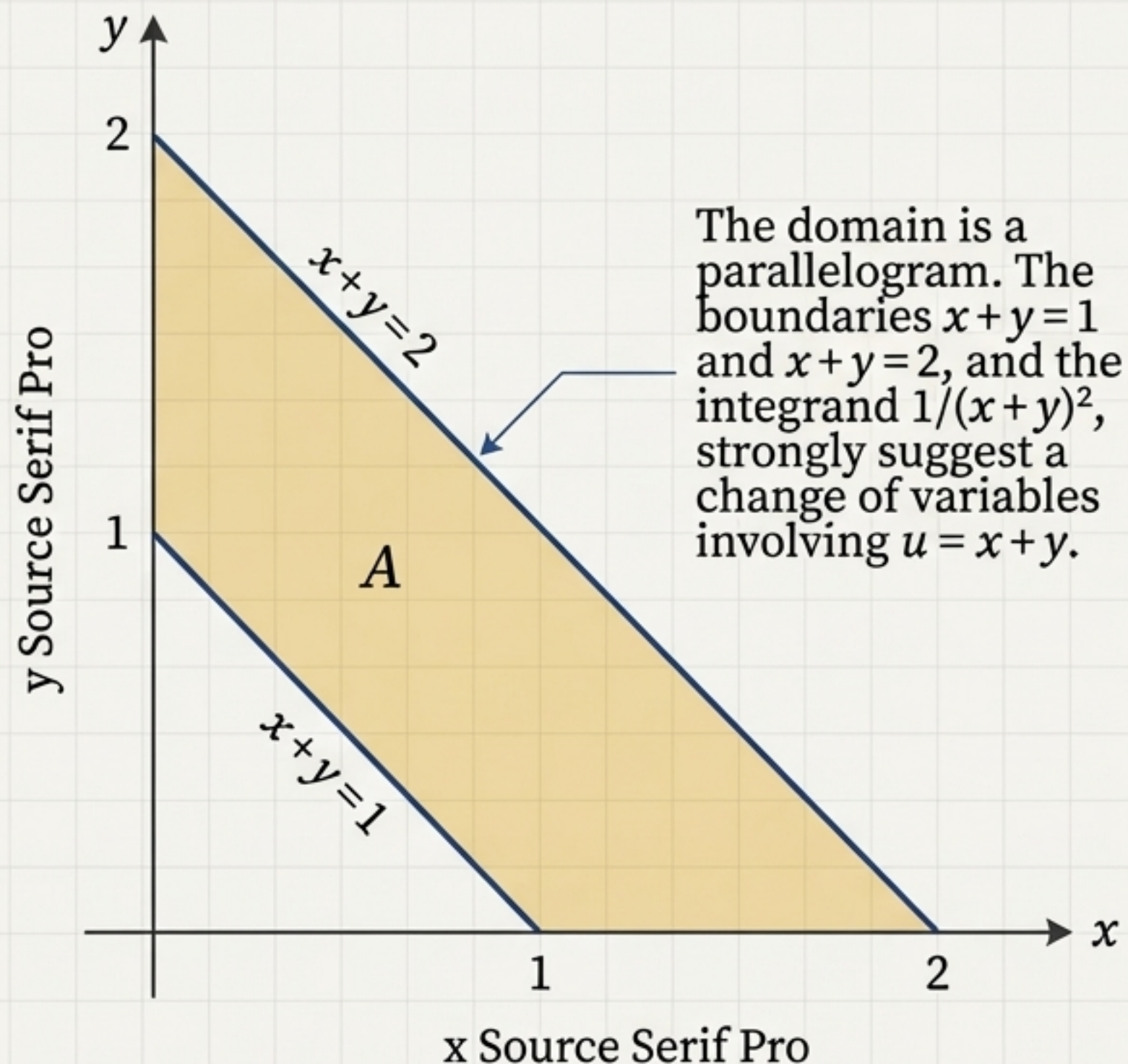
- Draw the set A.



Strategic Insight:

For double integrals, the challenge is rarely the integrand itself—it's **the domain**. The first and most critical step is to accurately **visualize and sketch** the domain A. The **geometry of the domain** dictates the entire solution strategy, especially whether a change of variables is needed.

The Blueprint & Build: Visualize and Transform



The Transformation

Define New Variables: Let $u = x+y$ and $v = x-y$.

Transform the Bounds:

- $1 \leq x+y \leq 2 \rightarrow 1 \leq u \leq 2$.
- $x \geq 0 \rightarrow \frac{u+v}{2} \geq 0 \rightarrow v \geq -u$.
- $y \geq 0 \rightarrow \frac{u-v}{2} \geq 0 \rightarrow v \leq u$.

Calculate the Jacobian $|J|$:

- $x = \frac{u+v}{2}, y = \frac{u-v}{2}$.
- The Jacobian determinant is $\left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \left| \frac{1}{2}(-\frac{1}{2}) - \frac{1}{2}\frac{1}{2} \right| = \left| -\frac{1}{2} \right| = \frac{1}{2}$.

The New Integral: $\iint_A f(x,y) dx dy = \iint_{A'} f(u,v) |J| du dv$.

$$\int_{u=1}^2 \int_{v=-u}^u \frac{1}{u^2} \frac{1}{2} dv du$$

The Build & Playbook: Integration and Strategy

The Final Calculation

$$\begin{aligned}\int_1^2 \int_{-u}^u \frac{1}{2u^2} dv du &= \frac{1}{2} \int_1^2 \frac{1}{u^2} [v]_{-u}^u du = \frac{1}{2} \int_1^2 \frac{1}{u^2} (u - (-u)) du = \frac{1}{2} \int_1^2 \frac{2u}{u^2} du \\ &= \int_1^2 \frac{1}{u} du = [\ln|u|]_1^2 = \ln(2) - \ln(1) = \boxed{\ln(2)}\end{aligned}$$

The Double Integral Playbook

1. **Sketch the Domain:** Always start by drawing the region of integration. This is non-negotiable.
2. **Assess the Geometry:** Do the boundaries or the integrand suggest a coordinate system change?
 - Circles/Arcs → Polar coordinates.
 - Tilted parallelograms / expressions like $x+y$ or $x-y$ → Linear transformation $u=...$, $v=...$
3. **Execute the Transformation:**
 - Define u and v .
 - Solve for x and y in terms of u and v .
 - Transform the integration bounds.
 - Calculate the absolute value of the Jacobian $|J|$.
4. **Evaluate the New Integral:** The transformed integral should be significantly simpler to solve.

Your Final Playbook: From Problems to Principles

This exam doesn't just test what you know; it tests how you think. Every problem is a signal, pointing to a core principle of analysis.



Identify the **Core Concept**.

(What is this problem *really* asking? Limit definition? Geometric series? Change of variables?)



Apply the **Right Tool**.

(Do I need L'Hôpital's rule? An identity? The Jacobian?)



Execute with **Precision**.

(Careful algebra, check your bounds, watch for details.)

You have the **theory** from your courses. This playbook gives you the strategy. Connect them, and you will not just solve the problems—you will master the exam.